Department of Computer Science and Engineering
University of Barishal

Final Examination

3rd Year 2nd Semester, Session: 2020-21

Course Code: CSE-3203 **Course Title: Theory of Computation**

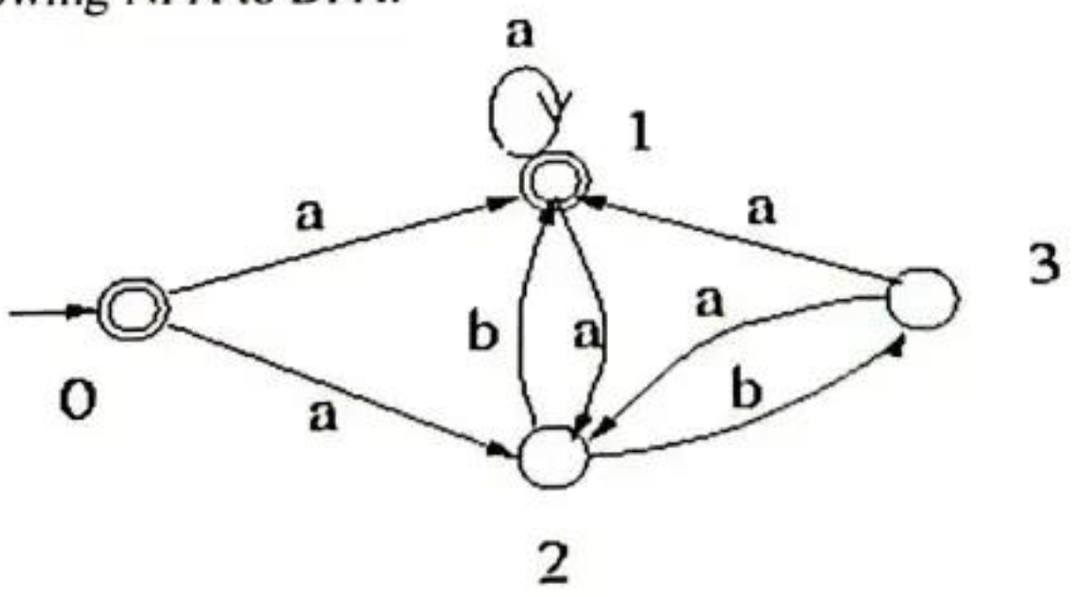
Total time: 3.00 hours **Total marks: 60.0**

(Note: Answer any five set of questions from the followings)

1. a) Define a Turing Machine and explain a pushdown automaton. 1+3
   b) Show that any language recognized by a multitape Turing machine is recursively enumerable. 6
   c) What do you know about context-free grammar's? 2

2. a) Provide definitions and examples for the following terms: 2+2+2
   i) Alphabet, ii) String, and iii) language
   b) Give a regular expression for the following language B over the alphabet {a, b}. 4
   $B = \{w \mid w \text{ does not contain the substring aaa}\}$.
   c) Under what circumstances is an expression referred to as a regular expression? 2

3. a) Use the pumping lemma to prove that the language $A = \{ 0^{2n} 1^{3n} 0^{n} \mid n \geq 0 \}$ is not context free. 5
   b) Convert the following NFA to DFA. 4



   c) State the Pumping Lemma for regular languages. 3

4. a) Summarize the principal closure properties for regular languages. 3
   b) Convert DFA's to regular expressions by eliminating states. 5
   c) Prove that if L is a regular language over alphabet Σ, then $\overline{L} = \Sigma^* - L$ is also a regular language. 4

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